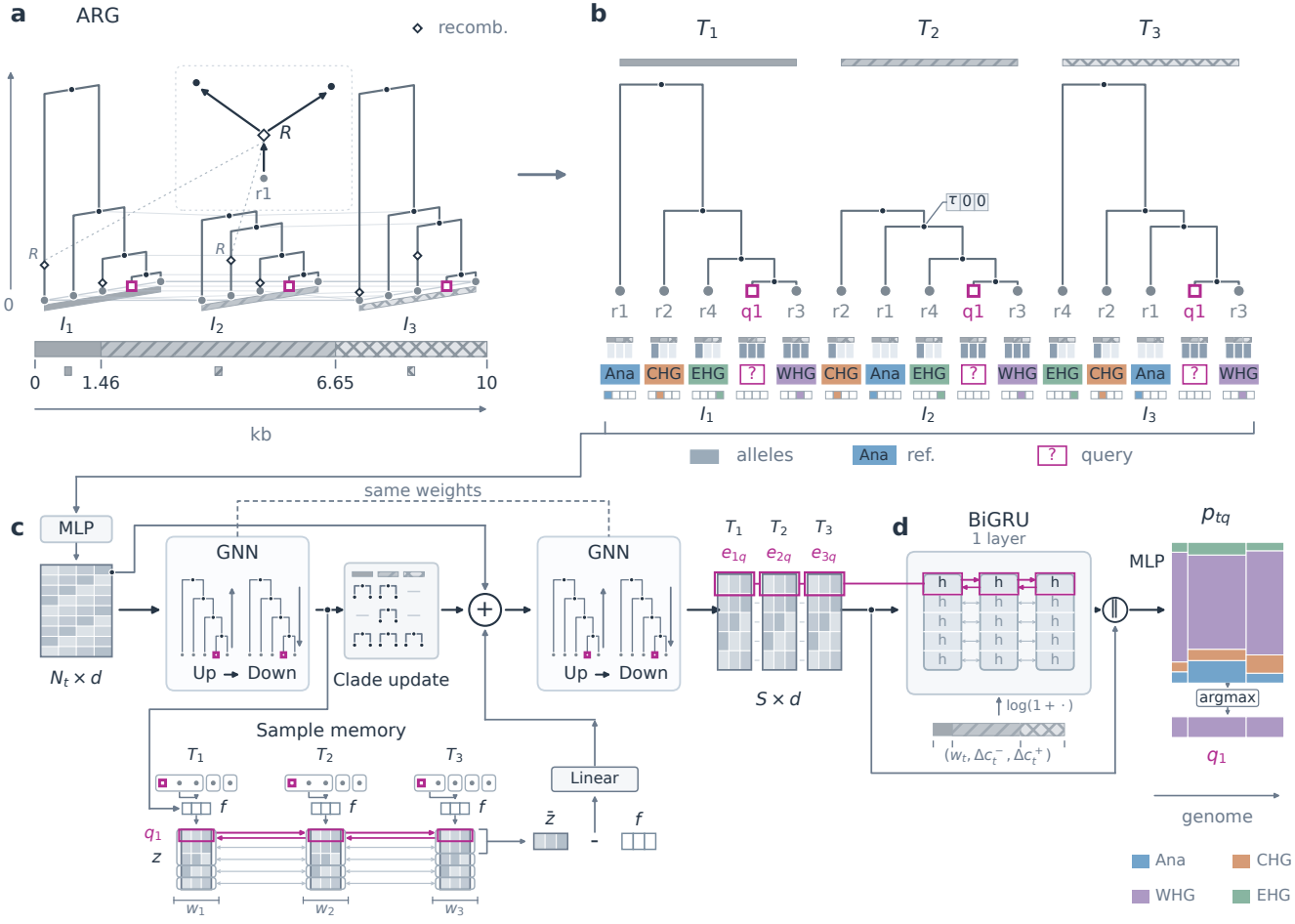

a
ARG
recomb.
time
0
R
r1
I_1
I_2
I_3
0
1.46
6.65
10
kb
b
T_1
T_2
T_3
τ 0 0
r1 r2 r4 q1 r3
Ana CHG EHG ? WHG
alleles
ref.
query
same weights
c
MLP
$N_t \times d$
GNN
Up → Down
Clade update
GNN
Up → Down
Sample memory
Linear
w_1
w_2
w_3
$S \times d$
d
BiGRU
1 layer
log(1 + ·)
MLP
p_{tq}
argmax
q_1
genome
Ana
CHG
WHG
EHG